Water Index Quality(WQI) can be reliably predicted using a stacking model with fewer, low-cost parameters than the traditional WQI formula

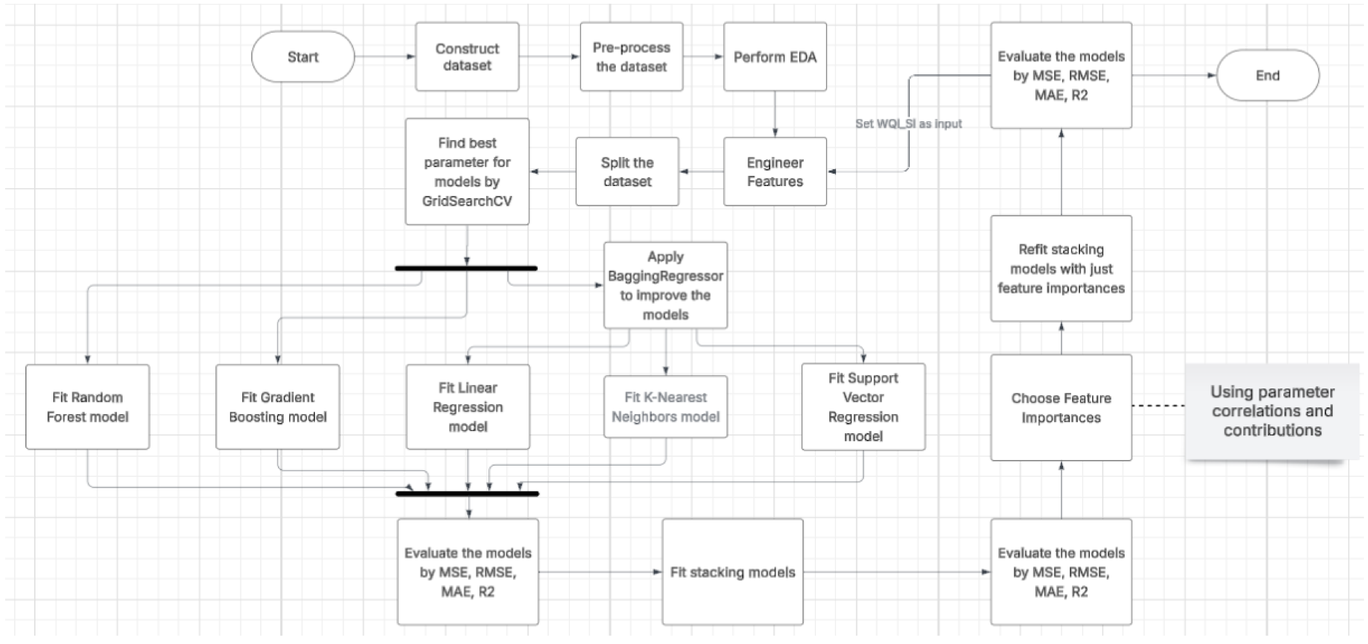
Title: Machine Learning-based Water Quality Index Prediction in the Quang Ninh Province, Vietnam

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BACKGROUND:

- Water quality management is critical, especially in Quang Ninh Province and Ha Long Bay (UNESCO site).
- Traditional WQI calculation is costly, time-consuming and risky due to extensive field sampling.
- AI-based WQI prediction is a recent trend due to its ability to use fewer parameters.

WORKFLOW



RESULT

Model	MSE	RMSE	MAE	R^2
Stacking model	4.399	2.097	1.066	0.97
Traditional WQI calculation	199.9	14.14	13.97	-0.56

- Removed As, Cd, NH_4,NO_3, BOD_5, Coliform
- Under reduced input conditions, the machine learning model demonstrated superior predictive performance relative to the conventional WQI calculation.

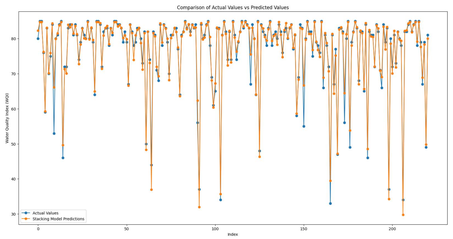
DICUSSION

- Instead of using raw parameters, WQI sub-indices were used as model inputs, resulting in better model performance.
- While previous studies often report that tree-based models perform best due to their robustness to outliers, our results show that after applying the transformed inputs, Linear Regression and Support Vector Regression achieved better performance.
- In practice, standardization performed better than robust standardization, which is consistent with findings reported in some existing studies.
- Using permutation importance for feature selection may remove critical information, making it harder to predict minority cases.

Descriptive Statistics of Water Quality Parameters

Parameter	Unit	Minimum	Maximum	Mean	Median	Std. Deviation
Temperature	Degree Celsius	7.4	39.4	24.222	23.6	4.843
pH	pH unit	4.59	8.64	6.685	6.64	0.444
DO	mg/L	1.08	9.7	6.703	6.72	0.997
COD	mg/L	1.5	306.4	15.604	11.3	23.158
BOD_5	mg/L	1	54.4	4.28	3.1	4.192
NO_3	mg/L	0.008	6.528	0.345	0.214	0.638
NO_2	mg/L	0.002	8.337	0.185	0.002	0.951
NH_4	mg/L	0.01	6.321	0.399	0.054	1.018
PO_4	mg/L	0.01	10.365	0.051	0.01	0.336
Coliform	MPN/100ml	3	25000	766.685	150	1885.311
EColi		3	400	14.118	3	35.718
As	mg/L	0	0.014	0.001	0	0.001
Hg	mg/L	0	0.006	0	0	0
Pb	mg/L	0	0.066	0.001	0.001	0.003
Cd	mg/L	0	0.011	0.001	0.001	0
CrVI		0.001	0.03	0.001	0.001	0.003





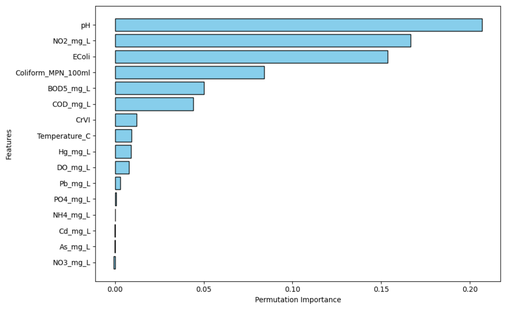
Direct Measurement Inputs

Before removing unimportant features

Model	MSE	RMSE	MAE	R^2
Random Forest	3.54	1.88	0.71	0.97
Gradient Boosting	3.43	1.85	0.75	0.97
Linear Regression	0.75	0.87	0.45	0.99
K-Nearest Neighbors	2.84	1.69	0.70	0.98
Support Vector Regression	0.73	0.85	0.40	0.99

Stacking Models Performance Metrics (5-Fold CV, Sorted by RMSE)

Base Models	Num Models	MSE	RMSE	MAE	R2	MSE_std	R2_std
RandomForest + LinearRegression + SVR + GradientBoosting	4	4.535	2.1295	1.0687	0.9645	1.8214	0.0114
LinearRegression + SVR + GradientBoosting	3	4.5732	2.1385	1.0182	0.9642	1.8129	0.0116
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	4.5748	2.1389	1.0171	0.9642	1.8566	0.0118
KNN + LinearRegression + SVR + GradientBoosting	4	4.5968	2.144	1.0249	0.964	1.8602	0.012
RandomForest + LinearRegression + GradientBoosting	3	4.7589	2.1815	0.9534	0.9629	2.0389	0.0125
LinearRegression + GradientBoosting	2	4.7775	2.1858	0.9539	0.9628	2.0173	0.0124
RandomForest + KNN + LinearRegression + GradientBoosting	4	4.9098	2.2158	0.9803	0.9615	1.9849	0.0127
RandomForest + SVR + GradientBoosting	3	4.934	2.2213	1.0814	0.9609	1.1461	0.0086
KNN + LinearRegression + GradientBoosting	3	4.9372	2.222	0.9832	0.9612	1.9614	0.0127
RandomForest + GradientBoosting	2	4.9661	2.2285	1.042	0.9606	1.2056	0.009
RandomForest + KNN + SVR + GradientBoosting	4	5.1182	2.2623	1.1002	0.9582	1.1528	0.01
SVR + GradientBoosting	2	5.1211	2.263	1.1175	0.9583	1.1776	0.0093
RandomForest + KNN + GradientBoosting	3	5.2342	2.2878	1.08	0.9582	1.256	0.0107
KNN + SVR + GradientBoosting	3	5.2901	2.3	1.1319	0.9577	1.1924	0.0107
KNN + GradientBoosting	2	5.4011	2.324	1.1055	0.9568	1.2739	0.0112
RandomForest + LinearRegression + SVR	3	6.5052	2.5505	1.3581	0.9488	2.2687	0.0154
RandomForest + KNN + LinearRegression + SVR	4	6.6277	2.5744	1.3629	0.9477	2.2851	0.0165
RandomForest + LinearRegression	2	7.377	2.7161	1.2727	0.9433	3.1524	0.0174
RandomForest + KNN + LinearRegression	3	7.9491	2.8194	1.3352	0.9375	2.6732	0.0173
RandomForest + SVR	2	8.1728	2.8588	1.5958	0.9349	1.4529	0.014
RandomForest + KNN + SVR	3	8.6784	2.9797	1.618	0.9285	1.6616	0.0194
RandomForest + KNN	2	9.7672	3.1284	1.6117	0.9214	1.6965	0.0187
KNN + LinearRegression + SVR	3	12.3008	3.5072	1.9633	0.9025	3.4155	0.0297
LinearRegression + SVR	2	12.989	3.604	2.0068	0.897	3.3412	0.0293
KNN + SVR	2	15.6979	3.9621	2.2835	0.8731	3.993	0.047
KNN + LinearRegression	2	20.9178	4.5736	2.5808	0.8377	5.5532	0.0271



After removing unimportant features

Stacking Models Performance Metrics (5-Fold CV, Sorted by RMSE)

Base Models	Num Models	MSE	RMSE	MAE	R2	MSE_std	R2_std
RandomForest + LinearRegression + SVR + GradientBoosting	4	4.535	2.1295	1.0687	0.9645	1.8214	0.0114
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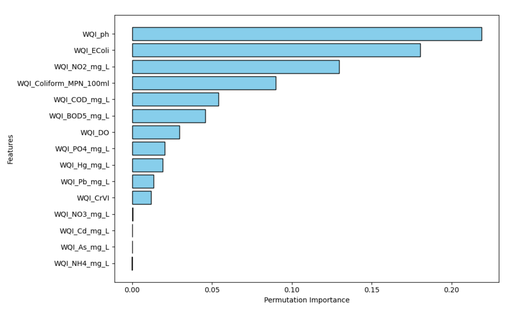
Feature Engineered Inputs (WQI_SI)

Before removing unimportant features

Model	MSE	RMSE	MAE	R^2
Random Forest	3.54	1.88	0.71	0.97
Gradient Boosting	3.43	1.85	0.75	0.97
Linear Regression	0.75	0.87	0.45	0.99
K-Nearest Neighbors	2.84	1.69	0.70	0.98
Support Vector Regression	0.73	0.85	0.40	0.99

Stacking Models Performance Metrics

Base Models	Num Models	MSE	RMSE	MAE	R2
LinearRegression + SVR	2	0.3035	0.5509	0.3548	0.9974
KNN + LinearRegression + SVR	3	0.3336	0.5776	0.3563	0.9972
RandomForest + LinearRegression + SVR	3	0.3437	0.5862	0.3575	0.9971
RandomForest + KNN + LinearRegression + SVR	4	0.3527	0.5939	0.3593	0.997
KNN + LinearRegression + SVR + GradientBoosting	4	0.4323	0.6575	0.3729	0.9963
LinearRegression + SVR + GradientBoosting	3	0.4359	0.6602	0.3724	0.9963
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	0.4407	0.6638	0.3814	0.9962
RandomForest + LinearRegression + SVR + GradientBoosting	4	0.4427	0.6654	0.3805	0.9962
KNN + LinearRegression	2	0.5464	0.7392	0.4057	0.9953
RandomForest + KNN + LinearRegression	3	0.5845	0.7645	0.4036	0.995
KNN + LinearRegression + GradientBoosting	3	0.6836	0.8268	0.4204	0.9942
RandomForest + KNN + LinearRegression + GradientBoosting	4	0.6897	0.8305	0.4284	0.9941
RandomForest + SVR	2	0.7514	0.8668	0.4734	0.9936
RandomForest + KNN + SVR	3	0.7653	0.8748	0.475	0.9935
RandomForest + LinearRegression	2	0.7975	0.893	0.4218	0.9932
KNN + SVR + GradientBoosting	3	0.8148	0.9027	0.4541	0.9931
SVR + GradientBoosting	2	0.8186	0.9048	0.4537	0.993
RandomForest + KNN + SVR + GradientBoosting	4	0.8326	0.9125	0.4686	0.9929
RandomForest + SVR + GradientBoosting	3	0.838	0.9154	0.4637	0.9929
KNN + SVR	2	0.9361	0.9675	0.4573	0.992
RandomForest + LinearRegression + GradientBoosting	3	0.9629	0.9813	0.4579	0.9918
LinearRegression + GradientBoosting	2	0.9901	0.9951	0.4696	0.9916
KNN + GradientBoosting	2	1.9233	1.3868	0.6348	0.9836
RandomForest + KNN + GradientBoosting	3	1.9273	1.3883	0.6469	0.9836
RandomForest + KNN	2	2.18	1.4765	0.7225	0.9814
RandomForest + GradientBoosting	2	3.4539	1.8585	0.7392	0.9705



After removing unimportant features

Stacking Models Performance Metrics (5-Fold CV)

Base Models	Num Models	MSE	RMSE	MAE	R2	MSE_std	R2_std
KNN + LinearRegression + GradientBoosting	3	4.3987	2.0973	1.0659	0.9651	1.2872	0.0103
LinearRegression + GradientBoosting	2	4.4058	2.099	1.0967	0.9653	1.2561	0.0093
RandomForest + KNN + LinearRegression + GradientBoosting	4	4.5922	2.1358	1.0798	0.9637	1.1769	0.011
RandomForest + LinearRegression + GradientBoosting	3	4.6542	2.1574	1.0976	0.9631	1.2767	0.011
LinearRegression + SVR + GradientBoosting	3	4.688	2.1652	1.0882	0.963	1.7052	0.0121
KNN + LinearRegression	2	4.7243	2.1736	1.0679	0.9623	1.4228	0.0115
KNN + LinearRegression + SVR + GradientBoosting	4	4.7551	2.1806	1.1	0.9625	1.736	0.0123
RandomForest + KNN + LinearRegression	3	4.781	2.1866	1.0928	0.962	1.0974	0.0096
RandomForest + LinearRegression + SVR + GradientBoosting	4	4.809	2.1929	1.0959	0.962	1.6766	0.0125
LinearRegression + SVR	2	4.8579	2.2041	1.094	0.9614	1.6936	0.0123
RandomForest + KNN + LinearRegression + SVR + GradientBoosting	5	4.8685	2.211	1.1121	0.9614	1.6848	0.0124
RandomForest + LinearRegression + SVR	3	4.906	2.2149	1.1041	0.9611	1.5709	0.0117
KNN + LinearRegression + SVR	3	4.9218	2.2185	1.0969	0.961	1.7973	0.013
RandomForest + LinearRegression	2	4.9545	2.2259	1.108	0.9607	1.14	0.0099
RandomForest + KNN + LinearRegression + SVR	4	5.001	2.2363	1.1157	0.9605	1.5937	0.0116
KNN + SVR + GradientBoosting	3	5.6972	2.3869	1.2198	0.9553	1.2948	0.0089
SVR + GradientBoosting	2	5.7384	2.3955	1.2172	0.9549	1.2176	0.0087
RandomForest + KNN + SVR + GradientBoosting	4	5.9224	2.4336	1.2279	0.9537	1.3106	0.0085
RandomForest + SVR + GradientBoosting	3	5.9287	2.4349	1.2313	0.9535	1.22	0.0084
KNN + GradientBoosting	2	6.386	2.527	1.2578	0.9494	1.032	0.0072
RandomForest + KNN + SVR	3	6.6278	2.5753	1.2745	0.9494	1.2013	0.0095
RandomForest + SVR	2	6.6598	2.5416	1.2891	0.949	1.1082	0.0094
KNN + SVR	2	6.6956	2.5487	1.2881	0.9483	1.2913	0.0118
RandomForest + KNN + GradientBoosting	3	6.7026	2.5889	1.277	0.9474	1.251	0.0081
RandomForest + GradientBoosting	2	7.008	2.6473	1.2903	0.9452	1.5225	0.0094
RandomForest + KNN	2	8.0043	2.8282	1.4233	0.9366	1.208	0.0114